

Solution Architecture

Date: 15 February 2025 **Team ID:** SWTID-2026-7464 **Project Name:** Script-Controlled ACL – Restrict Record Access Based on Field Value **Maximum Marks:** 4 Marks

Solution Architecture:

The proposed solution architecture bridges the record-access requirement with a script-controlled authorization mechanism. A user request is evaluated by an ACL script, which reads the configured record field value and applicable user context, then returns an allow or deny decision before protected record data is exposed.

Administrator Configures ACL rule Field + condition	ACL Configuration Access Control Rule Script / Condition	User / Client Requests protected record	Script-Controlled ACL Evaluate field value + user context	Record Data Protected table / fields
	↓	↓	↔	↑
Updates rule when requirement changes	Stores configured access condition	Record request + access context	Decision: ALLOW if condition matches DENY otherwise	Record returned only after authorization

Architecture Components and Responsibilities

Component	Responsibility in the proposed solution
User / Client	Initiates a request to view a protected record and provides the applicable authenticated user context.
Administrator / ACL Configuration	Defines the target record/table, field, expected value or condition, and access behavior.
Script-Controlled ACL	Evaluates the record field value and relevant access context at authorization time.
Record Data Store	Contains the protected record and the field value used by the ACL decision.
Access Decision	Returns ALLOW when the configured condition is satisfied; otherwise returns DENY.
Validation / Testing	Tests matching, non-matching, missing, empty, and unexpected field values before deployment.

Solution Flow

Step	Flow
1	User requests access to a protected record.
2	The access-control mechanism invokes the configured ACL script.
3	The script reads the relevant record field value and applicable user/access context.
4	The configured condition is evaluated.
5	If the condition matches, access is allowed and the requested record can be returned; otherwise access is denied.
6	Administrators can test and maintain the rule as field-based access requirements change.

Template alignment: The uploaded template defines Solution Architecture as the bridge between business problems and technology solutions and asks the architecture to describe structure, characteristics, behavior, features, development phases, requirements, and delivery specifications. This document applies that framing to the field-value ACL use case. ■filecite■turn7file0■L8-L18■